

FUNCTIONS AND LIMITS MCQS

→ Aptitude
→ Book

<https://youtu.be/ar19yZllh3l>

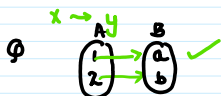


→ no

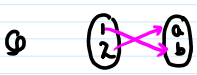
→ Domain → Input

→ Range → Output

Relation → Mapping



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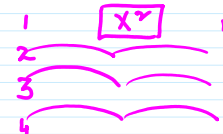
online

[A C] on R

0303-2765025

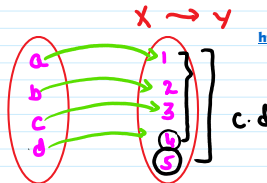
Butter

AC, For, h, WL, SB, SB

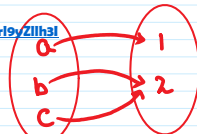


→ TYPES OF FUNCTIONS:

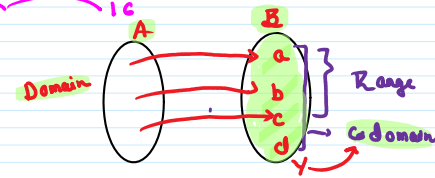
① Injective Function (1 to 1)



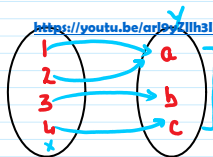
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MI = 50✓
SI = 100X



② Surjective (Onto)



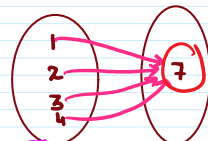
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$C \cdot d \Rightarrow R$ ✓

$R = C \cdot d$

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5 Constant

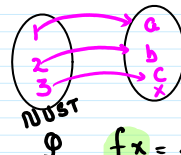


1 → 1
2 → 2
3 → 3
4 → 4
5 → 5

③ Into Function $C \cdot d \neq R$ X

④ Bijective → 1 to 1, onto
 $1 \rightarrow 1$
 $C \cdot d = R$

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→ Inverse Of A Function

① $f(x) = 3x + 5$

② $y = 3x + 5$

③ $x = f^{-1}y$
 $y = x$

$x \Rightarrow y - 5 = 3x$
 $x = \frac{y-5}{3} \rightarrow f^{-1}x = \frac{x-5}{3}$

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$f^{-1}x = \frac{-(3+6x)}{x-1}$

① $y = \frac{x-3}{x+6}$

② $yx + 6y = x - 3$

$4x - x = -3 - 6y$

$x(y-1) = -(3+6y)$

$x = \frac{-(3+6y)}{y-1}$

→ Composite Function

fog, gof,

$f(x) = x^2 + 2$ ✓ $g(x) = x + 5$

fog ⇒ $f(g(x)) = f(x+5) = (x+5)^2 + 2$

$f(g(x)) = (4x+3)^2 + 2$

$x^2 + 10x + 25 + 2$

$x^2 + 10x + 27$

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$16x^2 + 24x + 9 + 2$

$16x^2 + 24x + 11$

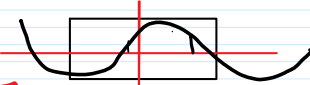
$g(f(g(x))) = (16x^2 + 24x + 11) + 5 = 16x^2 + 24x + 16$

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$$g(x) = 16x^2 + 24x + 11$$

$$g(f(x)) = (6x^2 + 24x + 11) + 5 = 16x^2 + 24x + 16$$

Even Function / Odd Function



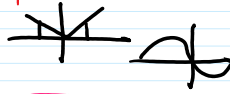
$$\sin(-\theta) = -\sin \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cos(-\theta) = \cos \theta$$

$$f(-x) \Rightarrow f(x)$$

Even



$$f(-x) = -f(x)$$

Odd

$$f(-x) = -f(x)$$

$$g(x^2 + \cos x) \rightarrow f(x)$$

$$f(x) = x^2 + \cos x \rightarrow f(x)$$

$$f(x) = x^3 + \sin x$$

$$f(-x) = (-x)^3 + \sin(-x)$$

$$= -x^3 - \sin x$$

$$f(-x) = -f(x)$$

LIMIT

Behaviour → Point

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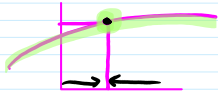
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$$0^{\infty} \Rightarrow \infty$$

$$\frac{0}{0} = 0$$

$$\frac{\infty}{\infty} = \infty$$

$$\frac{2^{999999}}{2^{999999}} = 1$$



$$L \cdot HL = R \cdot HL$$

$$\lim_{x \rightarrow a} f(x)$$

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots \rightarrow 0$$

$$1, 2, 3, 4, \dots \rightarrow \infty$$

function → limit ✓
Convergent ✓

function → limit +
Divergent

Convergent

Divergent

Properties

$$\frac{d}{dx} c = 0$$

$$\int c dx = cx + c$$

$$\lim_{x \rightarrow a} c = c$$

$$\lim_{x \rightarrow a} c \cdot x = c \cdot \lim_{x \rightarrow a} x$$

$$\lim_{x \rightarrow a} x^n = \left[\lim_{x \rightarrow a} x \right]^n$$

$$\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

L · H · R

$$\lim_{x \rightarrow 0} \frac{4x^2 - 3x}{2x}$$

L · H · R

$$\frac{0}{0}$$

$$\frac{0}{0} = \infty \times$$

$$\frac{-3}{2}$$

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Med, Fast

$$\lim_{x \rightarrow 0} \frac{1 - \cos 7x}{1 - \cos 5x} = \frac{1 - 1}{1 - 1} = \frac{0}{0}$$

$$\sin 0 = 0$$

$$\cos 0 = 1$$

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$$\frac{0 + \sin 7x (7)}{0 + \sin 5x (5)} = \frac{7 \sin 7x}{5 \sin 5x}$$

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$$\lim_{x \rightarrow 0} \frac{7 \cos 7x}{5 \cos 5x} = \frac{7}{5}$$

$$\lim_{x \rightarrow 2} \frac{3x^2 - 9}{2} = \frac{3(4) - 9}{2} = \frac{3}{2}$$

$$\lim_{x \rightarrow 2} \frac{3x^2 - 9}{2} = \frac{3}{2}$$

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 7x + 4}{2x^2 + 4x + 7}$$

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 7x + 4}{2x^2 + 4x + 7} = \frac{3}{2}$$

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$$\rightarrow \lim_{x \rightarrow \infty} \frac{3x^2 + 7x + 4}{2x^2 + 4x + 7} \rightarrow \frac{3}{2} \checkmark$$

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$$\lim_{x \rightarrow \infty} \frac{3x^4 - 1}{4x^3 + 7x} \rightarrow \frac{3}{4} \rightarrow \infty$$

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$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \checkmark$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow \infty} \left[1 + \frac{1}{x} \right]^x = e \checkmark$$

$$\lim_{x \rightarrow 0} \left[1 + \frac{1}{x} \right]^x = e \checkmark$$

$$\lim_{x \rightarrow \infty} \frac{\ln(e^x + 1)}{x} = 1 \quad \rightarrow \quad \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} \Rightarrow n a^{n-1}$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$